

JAMES EDWARD MITCHELL

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PROFESSIONAL SUMMARY

I'm a Research Engineer with a passion for exploring emerging technologies and solving complex technical problems. Based in Birmingham, I've spent the past year diving deep into machine learning systems and distributed computing. I enjoy writing clean code and collaborating with teams to turn research concepts into working prototypes. Outside of formal projects, I contribute to open source communities and maintain several personal projects focused on data processing and system optimization.

WORK EXPERIENCE

Research Engineer

DataLabs AI

www.datalabs.io

Birmingham, United Kingdom

September 2023 - Present

Developed and evaluated machine learning models for time series forecasting using Python, PyTorch, and TensorFlow on datasets containing over 5 million records.

Implemented data preprocessing pipelines using Pandas and NumPy to clean and transform raw sensor data from IoT devices, reducing data quality issues by 35 percent.

Collaborated with senior researchers to design experiment frameworks and document findings in technical reports and peer-reviewed publications.

Maintained version control using Git and participated in code reviews, ensuring code quality and knowledge sharing across the research team.

Junior Data Engineer

Insight Systems Ltd

www.insightsystems.co.uk

Birmingham, United Kingdom

April 2023 - August 2023

Built ETL pipelines using Python and SQL to extract data from multiple sources and load into PostgreSQL databases for analytical purposes.

Optimized database queries that reduced average query execution time from 45 seconds to 8 seconds, improving dashboard refresh rates.

Assisted in designing and implementing monitoring systems for data quality checks using Apache Airflow and custom Python scripts.

Documented technical specifications and created runbooks for data pipeline maintenance and troubleshooting.

Software Development Intern

TechVenture Solutions

www.techventure.net

Birmingham, United Kingdom

June 2022 - March 2023

Developed backend services using Python Flask and RESTful APIs to support a web-based analytics platform used by 200 plus customers.

Wrote unit tests and integration tests using pytest, achieving 78 percent code coverage across the API layer.

Debugged production issues and implemented fixes in collaboration with the DevOps team, reducing critical incidents by 40 percent during my tenure.

Participated in agile ceremonies including daily standups, sprint planning, and retrospectives, learning industry development practices.

EDUCATION

Bachelor of Science in Computer Science

University of Birmingham

Birmingham, United Kingdom

Graduated: June 2022

Grade: 2:1 (Upper Second Class Honours)

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript

Data Processing: Pandas, NumPy, Apache Spark

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost

Databases: PostgreSQL, MongoDB, SQLite

Tools and Platforms: Git, Docker, Apache Airflow, Linux, AWS

Data Visualization: Matplotlib, Seaborn, Plotly

Version Control: Git, GitHub

GITHUB REPOSITORIES

Time Series Anomaly Detection

github.com/jmitchell92/ts-anomaly-detection

Python implementation of anomaly detection algorithms for sensor data using isolation forests and autoencoders, with sample datasets and comprehensive documentation.

ETL Pipeline Framework

github.com/jmitchell92/etl-framework

Lightweight Python library for building modular ETL pipelines with support for multiple data sources and transformation plugins.

ML Experiment Tracker

github.com/jmitchell92/ml-experiment-tracker

Command line tool for tracking machine learning experiments, hyperparameters, and results, with CSV export functionality.

Data Quality Dashboard

github.com/jmitchell92/dq-dashboard

Flask based web application for monitoring data quality metrics across multiple databases with real time alerts.